ALLSEIS-4CLF Broadband Seismometer

Built for High Resolution, High Efficiency, and High Reliability Seismic Acquisition with Less Operational Cost and Environmental Impact

General Features

Sensor	Orthogonal 3-component broadband force-feedback geophone (0.2Hz/1Hz) + 10Hz high-sensitivity vertical geophone
Digitalization	4-channel 24-bit Δ-Σ ADC with GNSS disciplined clock
Storage Capacity	32G/64G (2ms sampling, four channels 45/90 days continuous recording)
Battery Life	Internal 130WH lithium-ion battery pack , Over 800 hours (33days) continuous recording , Support expansion of external Li-Ion battery unit or solar panel power supply
Timing Accuracy	±10μs (continuous GNSS disciplined) ±1ms (within 1 hour without GNSS signal) (External active GNSS antenna optional)
Working Mode	Autonomy with wireless on-site QC inspection
LED Status Indicator	Nodal health, GNSS synchronization status, Wireless data harvesting status
Data Harvest	Infield: Wireless Download Unit Indoor: Centralized Data Download Rack
Weight	2.9kg
Size	Diameter 14cm, Height 17cm
Operating Temperature	-40°C ~ +70°C
Waterproof	3 meters of water depth, 48 hours without leakage

Acquisition Specifications

ADC Resolution	24-bit high resolution
Sample Rates	0.5ms、1ms、2ms、4ms、10ms
Gain Accuracy	0.1%
Full Scale Input Signal	±2.5Vp-p
Instantaneous DR	125dB @ 2ms (typical value)
Equivalent Input Noise	1μV @ 2ms (typical value)
Anti-alias Filter	Linear filter, 86.6% of Nyquist -3dB bandwidth
Common Mode Rejection	>95dB
Stopband Attenuation	>105dB @Nyquist

Product Highlights

- Four-component broadband seismometer for passive and active-source seismic acquisition;
- Orthogonal 3-component broadband active force feedback geophone with negative impedance compensation （A series: 0.2Hz/B series: 1Hz） + 10Hz high-sensitivity vertical geophone （18Hz/60Hz/100Hz optional） ;
- Four channels of 24-bit Δ-Σ ADC , Configurable Preamplifier gain and sample rate;
- High sensitive GNSS module equipped with active high-gain patch antenna supports Beidou , GPS and GLONASS,;
- TNC external active GNSS antenna connector for deep planting and underwater deployments;
- BLE module together with visual LEDs supports onsite configuration, proximity, and real-time data transmission;
- Intelligent battery management unit supports automatic switchover between internal battery and external 12V DC power supply;
- High strength plastic enclosure with waterproof (IP68) and corrosion resistant;
- Compact volume (14D x 17h cm) and light weight (2.9kg) for convenient field transportation and deployment.

0.2Hz Low Frequency Geophone

Bandwidth	0.2Hz (5s) ~ 240Hz
Sensitivity	200V/m/s (±10%)
Tilt	±5°
Amplitude Consistency	< 5%
Phase Consistency	< 0.1s
Lateral Suppression	< 0.1%

1Hz Low Frequency Geophone

Bandwidth	1Hz (1s) ~ 240Hz
Sensitivity	200V/m/s (±10%)
Tilt	±5°
Amplitude Consistency	< 5%
Phase Consistency	< 0.1s
Lateral Suppression	< 0.1%

GDST 中科深源

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External Rechargeable Battery Unit

Capacity	195WHr (Nominal Capacity)
Charging Time	12 hours
Charging Temperature	0°C ~ +40°C
Discharging Temperature	-40°C ~ +70°C
Size	11.4 x 11.4 x 8.2cm
Weight	1.2kg

4G Remote Transmission Unit

Protocol Support	• GSM/GPRS/EDGE • WCDMA R99/Rel9DC-HSDPA+ • CDMA2000@1x, 1xEV-DOrA • LTE Cat4
Power Consumption	Data transmission state：800mW Online standby state：10mW
Battery Capacity	195WH(Nominal Capacity)
Weight	1.2kg
Size	11.4 x 11.4 x 8.2cm
Operating Temperature	-40°C ~ +70°C

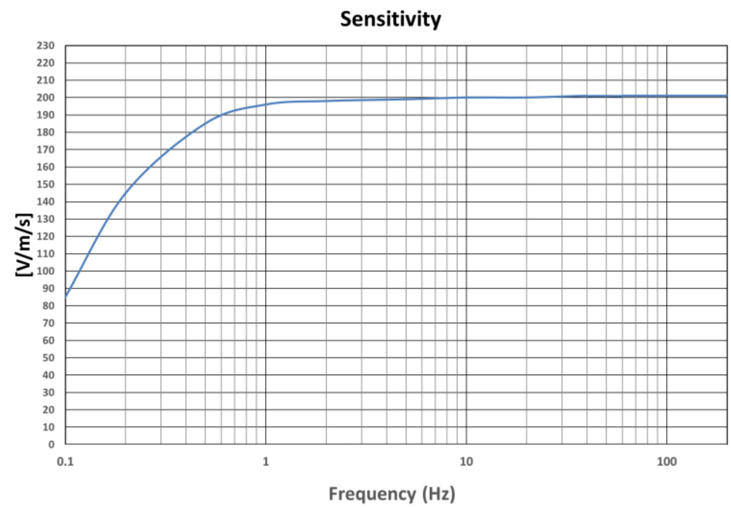
Stackable Charging Unit

Slots	12
Charging Time	8 hours
Charging Power	350W
External Dimensions (W x H x D)	48.3cm x 4.45cm x 45.5cm
Operating Temperature	0°C ~ +40°C

Stackable Data Download Unit

Slots	9
Slot Download Rate	600MB/min.
Full Download Rate	5.4GB/min.
External Dimensions (W x H x D)	48.3cm x 4.45cm x 45.5cm
Operating Temperature	0°C ~ +40°C

*The width and thickness of the stacked charging unit and data download unit meet the installation requirements of a standard 19-inch rack.



0.2Hz Low Frequency Geophone Response Curve

Stackable Data Download Unit（1U）

Stackable Charging Unit（1U）

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